

Aluminum 6010-T4

Categories: [Metal](#); [Nonferrous Metal](#); [Aluminum Alloy](#); [6000 Series Aluminum Alloy](#)

Material Notes: Data points with the AA note have been provided by the Aluminum Association, Inc. and are NOT FOR DESIGN.

Composition Notes:

Composition information provided by the Aluminum Association and is not for design.

Key Words: Aluminum 6010-T4; UNS A96010; Aluminium 6010-T4; AA6010-T4; Al6010-T4

Vendors: No vendors are listed for this material. Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	Metric	English	Comments
Density	2.71 g/cc	0.0979 lb/in ³	AA; Typical

Mechanical Properties	Metric	English	Comments
Hardness, Brinell	78	78	500 kg load with 10 mm ball
Hardness, Knoop	101	101	Converted from Brinell Hardness Value
Hardness, Vickers	88	88	Converted from Brinell Hardness Value
Tensile Strength, Ultimate	290 MPa	42100 psi	
Tensile Strength, Yield	170 MPa	24700 psi	
Elongation at Break	24 %	24 %	In 5 cm; Sample 1.6 mm thick
Modulus of Elasticity	69.0 GPa	10000 ksi	Average of Tension and Compression. In Al alloys, the compressive modulus is typically 2% greater than the tensile modulus. Estimated from trends in similar Al alloys.
Poissons Ratio	0.33	0.33	Estimated from trends in similar Al alloys.
Shear Modulus	26.0 GPa	3770 ksi	Estimated from similar Al alloys.
Shear Strength	195 MPa	28300 psi	

Electrical Properties	Metric	English	Comments
Electrical Resistivity	0.00000440 ohm-cm	0.00000440 ohm-cm	

Thermal Properties	Metric	English	Comments
CTE, linear 	21.5 µm/m-°C	11.9 µin/in-°F	
	@Temperature -50.0 - 20.0 °C	@Temperature -58.0 - 68.0 °F	
	23.2 µm/m-°C	12.9 µin/in-°F	
	@Temperature 20.0 - 100 °C	@Temperature 68.0 - 212 °F	
	24.1 µm/m-°C	13.4 µin/in-°F	
	@Temperature 20.0 - 200 °C	@Temperature 68.0 - 392 °F	
	24.8 µm/m-°C	13.8 µin/in-°F	average
	@Temperature 20.0 - 300 °C	@Temperature 68.0 - 572 °F	
	25.1 µm/m-°C	13.9 µin/in-°F	
	@Temperature 20.0 - 300 °C	@Temperature 68.0 - 572 °F	
Specific Heat Capacity	0.890 J/g-°C	0.213 BTU/lb-°F	Estimated from trends in similar Al alloys.
Thermal Conductivity	151 W/m-K	1050 BTU-in/hr-ft ² -°F	
Melting Point	585 - 649 °C	1090 - 1200 °F	

Solidus	585 °C	1090 °F
Liquidus	649 °C	1200 °F

Processing Properties	Metric	English	Comments
Annealing Temperature	413 °C	775 °F	
Solution Temperature	566 °C	1050 °F	
Aging Temperature	177 °C	350 °F	

Component Elements Properties	Metric	English	Comments
Aluminum, Al	95.3 - 98.3 %	95.3 - 98.3 %	As remainder
Chromium, Cr	<= 0.10 %	<= 0.10 %	
Copper, Cu	0.15 - 0.60 %	0.15 - 0.60 %	
Iron, Fe	<= 0.50 %	<= 0.50 %	
Magnesium, Mg	0.60 - 1.0 %	0.60 - 1.0 %	
Manganese, Mn	0.20 - 0.80 %	0.20 - 0.80 %	
Other, each	<= 0.05 %	<= 0.05 %	
Other, total	<= 0.15 %	<= 0.15 %	
Silicon, Si	0.80 - 1.2 %	0.80 - 1.2 %	
Titanium, Ti	<= 0.10 %	<= 0.10 %	
Zinc, Zn	<= 0.25 %	<= 0.25 %	

[References](#) for this datasheet.

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.